🌍 MOTION UNDER GRAVITY — FULL MIND MAP

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📝 Topics in this chapter:

1. From Brahmagupta to Newton — Development of Gravitation Ideas

- Brahmagupta (7th century) ne Brahmasphuta Siddhanta mein objects ke Earth ki taraf gravitate hone ka idea diya; source ise Newton se pehle ka anticipation maanta hai. *[Q1]*
- Isaac Newton ne 1687 mein universal law of gravitation publish kiya: har do masses ek-doosre ko attract karte hain. *[Q2, Q3]*

2. Gravity in Space — Microgravity, Inertia & Orbital Free Fall

- Orbit mein gravity absent nahi hoti; astronaut aur spacecraft dono continuous free fall mein saath accelerate karte hain, isliye support/normal force bahut kam hota hai aur apparent weightlessness feel hoti hai. *[Q4, Q24]*
- Orbiting spaceship se released apple spacecraft ki orbital velocity share karta hai, isliye woh roughly usi orbit mein spacecraft ke saath move karta rahega. *[Q5]*
- Artificial satellite ko Earth ka gravitational attraction hi required centripetal acceleration deta hai; isi wajah se satellite continuously "fall" karte hue Earth ko miss karta rehta hai. *[Q27, Q28, Q29]*

> ⚠️ **EXAM TRAP — Q4 — "No Gravity in Space" is an Exam Simplification**
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> ▸ Low-Earth orbit mein gravity zero nahi hoti; ISS altitude par gravity still substantial hoti hai. Weightlessness ka real reason spacecraft + astronaut ka common orbital free fall hai.
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> ▸ Agar old objective source option "There is no gravity" deta hai, source key ko note karo, but conceptually "microgravity/apparent weightlessness" correct explanation hai.

3. Centre of Gravity & Stability — Why the Leaning Tower Stands

- Body tab tak stable reh sakti hai jab centre of gravity se drawn vertical line uske base of support ke andar fall kare. Leaning Tower of Pisa isi condition ke karan abhi standing hai. *[Q6]*

4. Universal Gravitation — Inverse-Square Law & Radius Effect on g

- Newtonian gravitation: $F = Gm_1m_2/r^2$. Distance 2r ho jaaye to force F/4 ho jata hai. *[Q7]*
- Surface gravity $g = GM/R^2$. Earth ka mass same rakhkar radius 1% shrink ho to $g \approx g/(0.99)^2 \approx 1.02g$, yani roughly 2% increase. *[Q21]*

5. Free Fall — Same g, Mass Independence & Air Resistance

- Near Earth, true gravitational weight mg approximately same reh sakta hai while falling, lekin free-fall mein apparent weight/support force zero hota hai. *[Q8]*
- Vacuum mein feather, rubber ball aur wooden ball same g se fall karte hain; same height se release hon to saath reach karenge. *[Q9]*

- Ideal/no-air-resistance condition mein wood, wax, iron ya 10 kg vs 1 kg objects ka fall time mass par depend nahi karta; same height se drop karne par saath pahunchte hain. *[Q10, Q12]*
- Air present ho to drag shape/area par depend karta hai. Same-weight iron aur wood balls ke source example mein lower relative drag wali iron ball earlier reach kar sakti hai; vacuum result dono ke liye same hota. *[Q11]*
- Moon ke near bhi freely falling different masses same local gravitational acceleration paate hain, isliye same initial conditions par kisi instant unki velocity same hoti hai. *[Q13]*
- Rest se free fall: $d = \frac{1}{2}gt^2$, so $d \propto t^2$; displacement-time graph parabolic hota hai. *[Q18]*
- Dropped ball ke liye $v = u + gt$. $u = 0$, $g = 9.8\ m/s^2$, $t = 3\ s \Rightarrow v = 29.4\ m/s$ downward. *[Q20]*

⚠ EXAM TRAP — Q8 — True Weight vs Apparent Weight in Free Fall

- ▸ Gravitational weight mg near Earth roughly same rehta hai, kyunki m aur g almost unchanged hain.
- ▸ Lekin freely falling body ka apparent weight (normal/support force) zero hota hai. Question wording “weight” ko kis sense mein use kar rahi hai, ye exam mein check karo.

6. Moon Gravity — Weight Changes, Mass Does Not

- Moon ki surface gravity Earth ki approximately 1/6 hai; isliye person ka weight Moon par 1/6 hota hai. “Moon has no gravity” galat hai. *[Q14]*
- Mass location se nahi badalta: Earth par 100 kg mass Moon par bhi 100 kg hi rahega; sirf $W = mg$ change hota hai. *[Q23]*

7. Variation of g & Weight on Earth — Equator, Poles & Rotation

- Earth oblate hai: polar radius smaller aur equatorial radius larger. $g \propto 1/R^2$ ke karan g/weight maximum poles par aur minimum equator par hota hai. *[Q15, Q16]*
- Earth ki rotation equator par maximum centrifugal effect create karti hai, jo effective weight reduce karta hai. Rotation speed badhe to equator par effective weight aur decrease karega. *[Q17]*

8. Earthquake Acceleration — Source-Specific Threshold

- Source Q22 “destructive earthquake” ko ground acceleration greater than 980 cm/s² ($\approx 9.8\ m/s^2 = 1g$) se associate karta hai. Is statement ko source-specific exam key ke roop mein yaad rakho. *[Q22]*

⚠ EXAM TRAP — Q22 — Destructive Earthquake Acceleration

- ▸ Source answer > 980 cm/s² (>1g) deta hai. Ye universal definition of “destructive earthquake” nahi hai; earthquake severity magnitude, duration, frequency content, site response aur structural vulnerability par bhi depend karti hai.
- ▸ Isliye is fact ko conceptual law ke bajay source-specific PYQ key ke roop mein treat karo.

9. Motion on a Smooth Inclined Plane under Gravity

- Smooth incline par plane ke along acceleration $a = g \sin\theta$. Rest se length l cover karne ke liye $l = \frac{1}{2} g \sin\theta \cdot t^2$. *[Q19]*
- If vertical height $h = l \sin\theta$, then $t = (1/\sin\theta)\sqrt{(2h/g)}$. *[Q19]*

10. Why We Do Not Feel Earth’s Uniform Motion

- Hum Earth ke saath same rotational/orbital motion share karte hain; constant smooth motion ko relative frame mein feel nahi karte because our velocity relative to Earth is nearly zero. *[Q25]*

⚠ EXAM TRAP — Q25 — Source Stem’s 4,400 km/h Figure

- ▸ Earth ki actual mean orbital speed around Sun roughly 29.8 km/s (~107,000 km/h) hoti hai; 4,400 km/h source stem ka inaccurate value hai.
- ▸ Question ka intended concept phir bhi valid hai: uniform shared motion ko hum directly feel nahi karte; acceleration/change in motion feel hota hai.

11. If Earth’s Gravity Suddenly Vanished

- Gravitational force vanish ho jaaye to object ka gravitational weight $W = mg$ zero ho jayega, lekin intrinsic mass unchanged rahega. *[Q26]*

12. Four Fundamental Forces — Gravity is the Weakest

- Four fundamental interactions: gravitational, electromagnetic, weak nuclear and strong nuclear. Inki relative strength mein gravity weakest aur strong nuclear force strongest hota hai. *[Q30]*

13. Lift & Apparent Weight — When You Feel Heavier

- Lift upward acceleration a se move kare to normal reaction/apparent weight $N = m(g + a)$, so person heavier feel karta hai. *[Q31]*
- Downward acceleration mein $N = m(g - a)$; free fall $a = g$ par $N = 0$, yani apparent weightlessness. *[Q31]*

14. Simple Pendulum — Period, Seasons, Swing & Oscillation

- Simple pendulum ka time period $T = 2\pi\sqrt{(l/g)}$; ideal small oscillations mein period primarily effective length l aur g par depend karta hai. *[Q32]*
- Summer mein thermal expansion se pendulum length badhti hai ⇒ T badhta hai ⇒ clock slow/loses time. Winter mein contraction se length ghatti hai ⇒ T ghatti hai ⇒ clock fast chal sakti hai. *[Q33, Q36]*
- Swing par girl khadi hoti hai to centre of gravity upward shift hota hai aur effective pendulum length decrease hoti hai; therefore period shorter ho jata hai. *[Q34]*
- Mean position par restoring acceleration zero aur velocity maximum hoti hai; extreme position par velocity zero but acceleration maximum hoti hai. Real pendulum ka amplitude air resistance/friction ke karan time ke saath decrease karta hai. *[Q35]*

15. Escape Velocity & Why the Moon Lacks a Dense Atmosphere

- Escape velocity kisi celestial body ke gravitational field se permanently escape karne ki minimum launch speed hai. Earth ke liye ≈ 11.2 km/s. *[Q37]*
- 8 km/s Earth escape velocity se kam hai, isliye vertically projected object escape nahi karega aur idealized exam context mein Earth par return karega. *[Q38]*
- Moon ki low gravity ke karan escape velocity low hai; gas molecules ki thermal/RMS speeds ke comparison mein retention weak hoti hai, isliye Moon dense permanent atmosphere retain nahi kar pata. *[Q39, Q40]*

16. Candle Flame in Microgravity — Shape vs “No Flame” Confusion

- Microgravity mein buoyancy-driven convection weak ho jati hai; oxygen diffusion-dominated burning ki wajah se candle flame elongated nahi, roughly spherical hoti hai. *[Q42]*
- Actual vacuum mein oxygen absent hone se ordinary candle burn nahi karegi; microgravity spacecraft mein oxygen available ho to flame exist kar sakti hai, bas uska shape/combustion behavior change hota hai. *[Q41, Q42]*

> ⚠ **EXAM TRAP — Q41–Q42 — Candle in Space: Vacuum ≠ Microgravity**
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> ▸ Microgravity with oxygen: candle flame ban sakti hai aur roughly spherical hoti hai because buoyant convection weak hoti hai. Q42 isi concept se consistent hai.
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> ▸ Vacuum: oxygen na hone par ordinary candle burn nahi karegi. Isliye “space mein gravity nahi, isliye flame nahi” scientifically oversimplified/inconsistent statement hai.